An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka

R26-DS-004



Project Proposal Report

Hewagama S.R – IT22896186

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Data Science

Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

Intelligent Tax Advisory Language Model: A Neuro-Symbolic Small Language Model (SLM) for Grounded and Traceable Tax Compliance in Sri Lanka

R26-DS-004

Hewagama S.R – IT22896186

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Data Science

Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

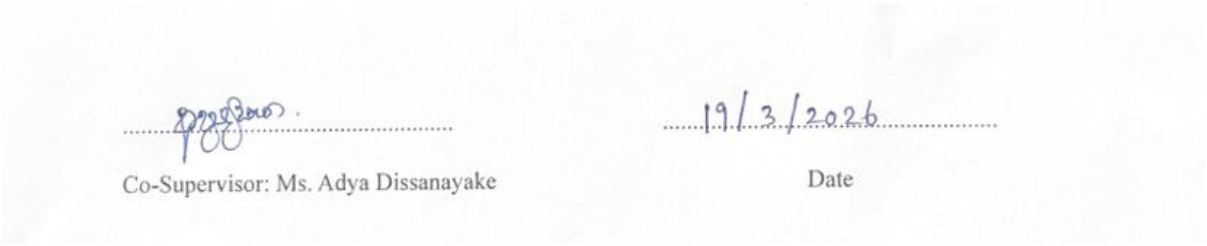
DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Hewagama S.R	IT22896186	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature



ABSTRACT

The **Intelligent Tax Advisory Language Model** addresses the critical "Trust Gap" in digital tax compliance by transitioning from probabilistic general-purpose Artificial Intelligence toward a deterministic, **Neuro-Symbolic architecture**. While Large Language Models (LLMs) frequently suffer from factual "hallucinations" and a lack of evidentiary traceability in the legal domain **[11]**, this research develops a domain-specific **Small Language Model (SLM)** fine-tuned on Sri Lankan tax regulations. The system integrates **Graph-Augmented Retrieval (GraphRAG)** to ground responses in verified Inland Revenue Act (IRA) sections and circulars, applying **Lex Specialis priority mapping** to resolve legal conflicts between current amendments and base regulations **[13]**. A core novelty is the **Agentic "Think Twice" loop**, which utilizes a **Symbolic Rule Engine** to self-correct generated advice before delivery, ensuring 100% compliance with hard-coded legal rules **[14]**. The output includes a visual **Proof Map**, transforming the "Black Box" of AI reasoning into a transparent, auditable justification for taxpayers and regulatory authorities alike **[15]**. **Experimental results demonstrate that this hybrid approach significantly reduces hallucination rates and improves procedural accuracy compared to general-purpose baseline models, providing a scalable framework for legally-compliant digital government services in Sri Lanka.**

ACKNOWLEDGEMENT

We would like to take this opportunity to express our sincere gratitude to our supervisor, **Dr. Lakmini Abeywardhana**, for her continuous support, guidance, and encouragement throughout the development of this research. Her valuable advice, insightful feedback, and academic direction have played a significant role in helping us shape and improve this study.

We are also deeply thankful to our co-supervisor, **Ms. Adya Dissanayake**, for her guidance, support, and thoughtful suggestions during the preparation of this proposal. Her observations and encouragement helped us strengthen the quality and clarity of our work.

We would also like to extend our appreciation to all lecturers and academic staff who have contributed to our learning journey and provided the knowledge and foundation needed to undertake this research successfully.

Finally, we are grateful to our team members, friends, and family for their constant encouragement, motivation, and support throughout this process.

This proposal would not have been possible without the guidance, support, and encouragement of all those who contributed to it.

Contents

DECLARATION....3

ACKNOWLEDGEMENT....5

LIST OF TABLES....9

LIST OF TABLES....9

LIST OF FIGURES....10

LIST OF ABBREVIATIONS....11

1. INTRODUCTION....12

1.1 Background and Context....12

1.1.1 Complexity of Modern Tax Systems....12

1.1.2 Digital Transformation in Tax Administration....12

1.1.3 Transition from General LLMs to Specialized SLMs....12

1.1.4 The "Black Box" and Jurisdictional Limitations....12

1.1.5 Need for Neuro-Symbolic Advisory Systems....12

1.1.6 Component Focus and System Alignment....13

1.2 Literature Review....13

1.2.1 The Hallucination and Traceability Challenge in Legal AI....13

1.2.2 Jurisdictional Failures and Local Law Limitations....14

1.2.3 Procedural Logic and Mathematical Reasoning Failures....14

1.2.4 The Explainability and "Black Box" Problem....14

1.3 RESEARCH GAP....15

1.3.1 Limitations in Current Tax Advisory Systems....15

1.3.2 Challenges of General LLMs in the Legal Domain....15

1.3.3 Need for Grounded and Traceable Advisory....15

1.3.4 Motivation for the Neuro-Symbolic SLM....16

1.3.5 Identified Research Gap of the Proposed Study....16

1.4 RESEARCH PROBLEM....17

1.4.1 Challenges in Accessing Tax Advisory Information 17

1.4.2 Limitations of "Black Box" and Rule-Based Systems 17

1.4.3 Reliability and "Lex Specialis" Logic Gaps 17

1.4.4 Research Problem Statement 17

2. OBJECTIVES 19

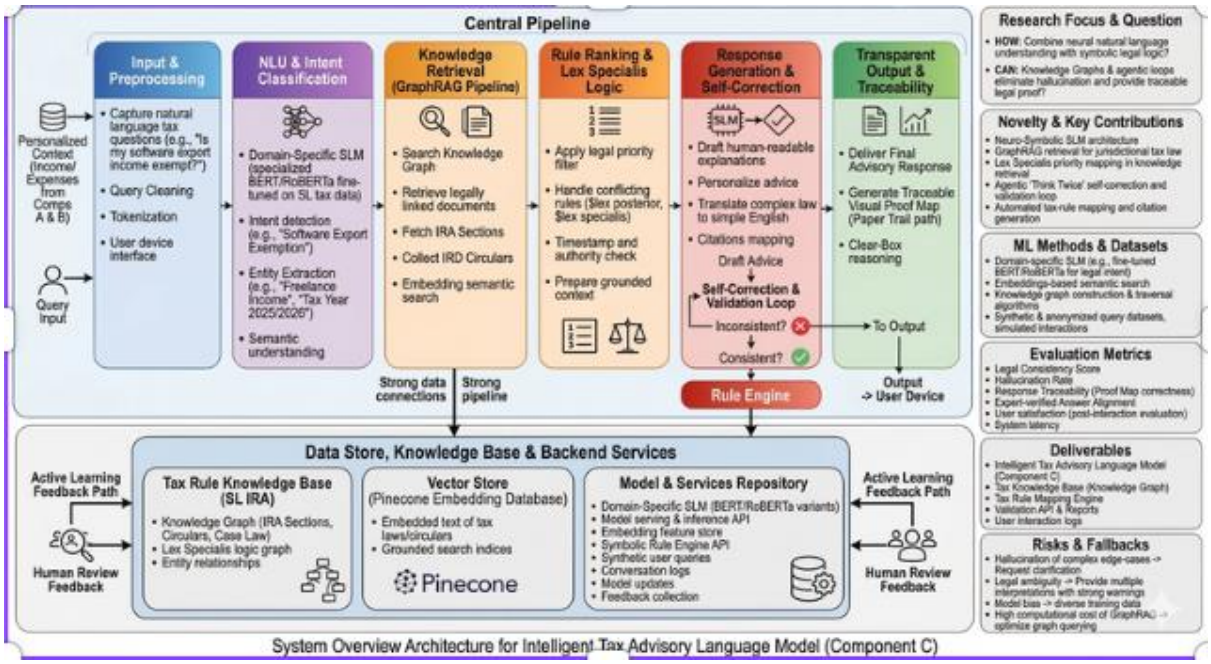
2.3 ALIGNMENT OF THE OBJECTIVES WITH THE RESEARCH PROBLEM 20

3. METHODOLOGY 21

3.11 Ethical Considerations 24

4. HIGH-LEVEL SYSTEM ARCHITECTURE 24

4.1 Overview of the Proposed Component Architecture 24



..... 24

4.2.1 Layer 1: Input & Contextual Preprocessing 25

4.2.2 Layer 2: The Cognitive Core (NLU & Knowledge Retrieval) 25

4.2.3 Layer 3: The Reasoning Engine (Generation & Validation) 25

4.2.4 Layer 4: The Transparent Output Layer (Traceability) 25

4.3 Internal Component Interactions 26

4.4 Data Flow Through the Component 26

4.5 Scalability and Future-Proofing Considerations 26

4.6 Security, Privacy, and Explainability.....26

5. USER REQUIREMENTS.....27

6. COMMERCIALIZATION PLAN 29

7. BUDGET AND JUSTIFICATION.....31

8. WORK BREAKDOWN STRUCTURE (WBS) 32

9. GANTT CHART.....33

REFERENCES.....34

LIST OF TABLES

LIST OF TABLES

Table 1 – Comparison of Existing Digital Tax Systems and the Proposed Intelligent Tax Advisory Language Model

Table 2 – Summary of Literature on AI-based Legal and Tax Advisory Systems (2025-2026)

Table 3 – Research Gap Analysis of Existing Tax Information and AI-based Advisory Systems

Table 4 – Alignment Between the Research Problem and Proposed Neuro-Symbolic Objectives

Table 5 – Tools and Platforms Used for the Development of the Neuro-Symbolic SLM Component

Table 6 – Estimated Budget for the Intelligent Tax Advisory Language Model Component

LIST OF FIGURES

Figure 1 – Overall Methodological Framework of the Intelligent Tax Advisory Language Model

Figure 2 – Neuro-Symbolic SLM Architecture for Grounded and Traceable Tax Advisory

Figure 3 – Work Breakdown Structure (WBS) and Key Deliverables

Figure 4 – Gantt Chart of the Proposed 12-Month Research Timeline

LIST OF ABBREVIATIONS

AI – Artificial Intelligence

API – Application Programming Interface

B2B – Business to Business

DL – Deep Learning

FAQ – Frequently Asked Questions

ICT – Information and Communication Technology

IRD – Inland Revenue Department

IT – Information Technology

LLM – Large Language Model

ML – Machine Learning

NLP – Natural Language Processing

QA – Question Answering

RAG – Retrieval-Augmented Generation

UI – User Interface

UX – User Experience

1. INTRODUCTION

1.1 Background and Context

1.1.1 Complexity of Modern Tax Systems Tax systems globally have reached unprecedented levels of complexity due to the rapid introduction of fiscal policies, regulatory amendments, and intricate deduction rules. In Sri Lanka, the Inland Revenue Act No. 24 of 2017 and its subsequent 2025 amendments require taxpayers to navigate dense legal terminology that often lacks clear interpretation for the average citizen. This complexity creates a significant "Interpretation Gap," where even minor errors in understanding procedural logic can lead to severe financial penalties. Improving accessibility to tax advisory through digital tools is essential for modern tax administration , [2], especially in jurisdictions like Sri Lanka, where low tax literacy remains a barrier to voluntary compliance, [16].

1.1.2 Digital Transformation in Tax Administration Governments have accelerated the adoption of digital technologies to modernize public services and improve information access. In Sri Lanka, platforms like RAMIS have digitalized return filing; however, these systems often remain functional rather than conversational, failing to provide the personalized guidance required by non-professionals, [4]. Conversational AI has emerged as a critical tool for bridging this gap, transforming government-citizen interaction by providing real-time support without requiring users to manually search through lengthy policy documents, [10].

1.1.3 Transition from General LLMs to Specialized SLMs While Large Language Models (LLMs) have demonstrated strong potential in natural language tasks, their application in high-stakes legal domains like taxation is hindered by significant risks. Recent studies indicate that general-purpose models often "hallucinate" or fabricate legal citations between 17% and 33% of the time. To mitigate these risks, this research moves toward **Small Language Models (SLMs)** that are fine-tuned on specific regulatory datasets. By integrating **Graph-Augmented Retrieval (GraphRAG)**, these models can be grounded in verified legal sources, ensuring that advisory responses are factual rather than probabilistic, [11].

1.1.4 The "Black Box" and Jurisdictional Limitations A primary limitation of existing tax AI systems is their "Black Box" nature, where the internal reasoning is opaque and lacks evidentiary traceability. Furthermore, general AI often fails to handle **Lex Specialis** (law priority) logic, failing to recognize when a specific 2025 IRD circular overrides a general provision from an older act. Current digital portals like RAMIS and rule-based chatbots lack the "procedural alignment" needed to guide a user through a multi-step legal calculation accurately. These gaps necessitate a transition toward more transparent architectures that provide clear legal justification, [15].

1.1.5 Need for Neuro-Symbolic Advisory Systems Given the legal risks of AI misinformation, there is a critical need for an **Intelligent Tax Advisory Language Model** that combines neural communication with symbolic legal logic. Such a **Neuro-Symbolic** approach allows the system to understand natural language while strictly adhering to hard-coded tax rules. By incorporating an **Agentic "Think Twice" loop**, the system can self-verify its advice against verified tax brackets before delivery, effectively eliminating

hallucinations. Furthermore, providing a visual **Proof Map** addresses the "Explainability Challenge" by showing users the exact "paper trail" from the law to their specific financial outcome.

1.1.6 Component Focus and System Alignment

Within the broader project framework, this component serves as the **Intelligent Tax Advisory Language Model**. It is designed as a Neuro-Symbolic SLM that transforms complex Sri Lankan tax regulations into grounded, traceable, and conversational guidance. The component integrates four critical novelties: **GraphRAG-based grounding**, **Lex Specialis priority logic**, **Adversarial Self-Correction**, and **Visual Proof Maps**. This alignment ensures that the advisory system not only simplifies interaction but maintains the high degree of regulatory consistency required for national tax compliance.

1.2 Literature Review

The rapid advancement of Artificial Intelligence (AI) technologies has significantly influenced the development of intelligent advisory systems across multiple domains, including finance, healthcare, and public administration. In particular, AI-driven systems have demonstrated strong potential for supporting decision-making processes by analyzing large volumes of information and delivering context-aware insights to users. In the domain of taxation, intelligent systems have begun to assist taxpayers by simplifying complex regulatory information, providing automated guidance, and improving access to financial advisory services.

Recent studies highlight the growing interest in applying conversational AI and natural language processing technologies within government services to improve citizen engagement and service delivery efficiency. Intelligent advisory systems can support users in understanding legal and financial regulations while reducing the burden on government service providers. This literature review examines existing research related to artificial intelligence in tax administration, conversational AI technologies, natural language processing techniques, retrieval-augmented knowledge architectures, and current AI-based tax advisory platforms. The review further identifies the limitations of existing approaches and establishes the need for reliable knowledge-grounded intelligent tax advisory systems [1], [3].

1.2.1 The Hallucination and Traceability Challenge in Legal AI

The primary barrier to deploying Large Language Models (LLMs) in the legal and tax domain is the phenomenon of "hallucination," where models generate plausible but factually incorrect legal citations. Recent evaluations of leading proprietary legal RAG tools (such as Lexis+ AI and Westlaw) have demonstrated hallucination rates between 17% and 33% **[11]**. Furthermore, standalone models often lack "procedural depth," failing to navigate the multi-stage workflows required for complex legal tasks, which often leads to fabricated citations and a lack of evidentiary traceability **[12]**.

1.2.2 Jurisdictional Failures and Local Law Limitations

Current research indicates a significant "Jurisdictional Gap" in general AI performance. While models perform adequately on broad federal laws, their accuracy drops sharply when tasked with local, place-based legal queries outside of major western data hubs **[13]**. In the context of Sri Lanka, general models struggle with the intricate syntax and specialized vocabulary of local statutes **[16]**. This justifies the need for domain-specific **Small Language Models (SLMs)** fine-tuned on local data to recognize jurisdictional hierarchies, such as the **Lex Specialis** principle where a specific circular overrides a general act.

1.2.3 Procedural Logic and Mathematical Reasoning Failures

Tax advisory requires strict adherence to mathematical and procedural steps defined by law. However, existing LLMs demonstrate poor performance in "Legal Mathematical Reasoning," with accuracy rates often falling below 15% **[14]**. These failures occur because neural models prioritize linguistic patterns over symbolic procedural logic. This research gap highlights the necessity for a **Neuro-Symbolic architecture** that aligns neural understanding with a hard-coded symbolic rule engine to ensure every calculation follows the mandatory legal sequence **[14]**.

1.2.4 The Explainability and "Black Box" Problem

In regulated fields like taxation, an answer without a "paper trail" is legally insufficient. Current generative architectures are largely viewed as "Black Boxes" because their internal decision-making processes are opaque and impossible to audit **[15]**. The "Explainability Challenge" identifies that as models become more capable, they become harder to interpret, creating a roadblock for government and legal adoption **[15]**. My research addresses this by implementing **Proof Maps** to transform these "Black Boxes" into transparent systems that provide clear, visual justification for every advisory output.

1.3 RESEARCH GAP

Despite the rapid advancement of generative artificial intelligence and the digital transformation of government services, significant limitations remain in existing tax information systems and AI-based advisory platforms. While current digital tax services provide automated assistance, they lack the **Neuro-Symbolic reasoning** required to interpret complex Sri Lankan tax regulations with 100% accuracy. General-purpose models often fail to provide **traceable legal justification**, leaving taxpayers at risk of non-compliance due to unverified AI outputs **[11]**, **[15]**.

1.3.1 Limitations in Current Tax Advisory Systems

Most existing tax information platforms in Sri Lanka primarily rely on static digital resources, such as official government FAQ sections and PDF documentation. While these provide raw regulatory data, they require users to manually navigate volumes of text to locate specific guidance. Rule-based chatbot systems have been introduced to improve accessibility, but these are limited by keyword-matching architectures that cannot interpret complex natural language or handle queries outside predefined templates **[3]**, **[4]**. Furthermore, these systems provide general information rather than personalized, grounded guidance, failing to adapt to the "mixed-income" profiles of modern taxpayers **[16]**.

1.3.2 Challenges of General LLMs in the Legal Domain

The application of general-purpose Large Language Models (LLMs) within the Sri Lankan tax domain introduces critical risks:

- **The Hallucination Gap:** Evaluations of leading AI legal tools demonstrate hallucination rates between 17% and 33%, where models fabricate legal citations that appear plausible but are factually incorrect **[11]**.
- **Jurisdictional Failure:** General AI models lack "place-based" legal grounding, leading to significant accuracy drops when applied to local jurisdictions like Sri Lanka, where they fail to recognize the hierarchical priority of local laws (**Lex Specialis**) **[13]**.
- **Mathematical & Procedural Failure:** Standalone LLMs demonstrate an average accuracy below 15% in legal mathematical reasoning because they lack the procedural alignment required to follow strict tax statutes **[14]**.

1.3.3 Need for Grounded and Traceable Advisory

Given the legal risks of AI misinformation, there is a growing need for an **Intelligent Tax Advisory Language Model** that moves beyond the "Black Box" nature of standard generative AI. Such a system must combine natural language understanding with **symbolic legal logic** to ensure every response is anchored to the Inland Revenue Act. By integrating

Graph-Augmented Retrieval (GraphRAG), the system can provide "Clear Box" transparency, offering users a visual **Proof Map** that links their financial data directly to the specific legal clause used to generate the advice **[12]**, **[15]**.

1.3.4 Motivation for the Neuro-Symbolic SLM

The limitations identified in existing platforms highlight the need for a hybrid solution. The proposed **Neuro-Symbolic Small Language Model (SLM)** is designed to address these challenges through:

1. **Domain-Specific Fine-Tuning:** An SLM specialized in Sri Lankan tax syntax to avoid the errors of generalized models **[16]**.
2. **Agentic "Think Twice" Loop:** An adversarial validation layer that tests generated advice against a hard-coded **Symbolic Rule Engine** to prevent hallucinations before delivery **[12]**.
3. **Legal Priority Mapping:** Utilizing **Lex Specialis logic** to ensure the system prioritizes 2025 amendments over older, generalized provisions **[13]**.

1.3.5 Identified Research Gap of the Proposed Study

Based on the analysis of current literature and existing systems, the primary research gap lies in the **absence of a grounded, traceable, and jurisdiction-aware conversational tax advisory system** for Sri Lanka.

- **First**, current systems fail to resolve legal conflicts between old and new regulations, lacking the **Lex Specialis** priority logic required for compliance.
- **Second**, most existing AI models lack **evidentiary traceability**, providing answers without the visual "paper trail" (Proof Maps) necessary for legal auditing **[15]**.
- **Third**, there is a lack of **procedural alignment** in AI-driven tax tools, leading to mathematical failures in complex income-tax calculations **[14]**.

To address this gap, the proposed study introduces a hybrid architecture that combines a **Small Language Model (SLM)** with **Graph-Augmented Retrieval (GraphRAG)** and **Agentic Self-Correction**, ensuring reliable, personalized, and fully traceable tax guidance **[11]**, **[12]**.

Static Content Gap: Existing government and accounting portals rely on static FAQs that cannot handle complex, multi-turn natural language queries.

The "Hallucination" Gap: General-purpose LLMs (like ChatGPT) frequently guess when faced with niche local laws, creating significant legal risks for users.

The "Black Box" Gap: Current digital tools provide answers without a "paper trail" or verifiable proof, making it impossible to audit the AI's logic.

Legal Conflict Blindness: Standard AI systems fail to recognize jurisdictional hierarchies, such as a **2025 IRD Circular** overriding a **2017 Base Act** (Lex Specialis)

1.4 RESEARCH PROBLEM

The complexity of modern taxation systems presents a significant barrier to economic participation and legal compliance. In Sri Lanka, the Inland Revenue Act No. 24 of 2017 and its subsequent 2025 amendments are characterized by dense legal terminology and intricate procedural logic that the average citizen cannot accurately interpret. While digital transformation has introduced portals like RAMIS, these systems are primarily functional repositories and lack the "Intelligent Reasoning" required for personalized advisory support.

1.4.1 Challenges in Accessing Tax Advisory Information

A major challenge is the "Interpretation Gap" caused by the legalistic nature of tax documentation. Ordinary taxpayers struggle to manually search through large volumes of unstructured data to identify relevant clauses for their specific financial situation. This inefficiency often leads to incorrect filings, missed deductions, or unintentional non-compliance.

1.4.2 Limitations of "Black Box" and Rule-Based Systems

Existing digital tax solutions typically rely on static FAQs or keyword-based rule engines. These systems lack the capability to handle multi-turn natural language queries or interpret complex financial contexts. Furthermore, when taxpayers turn to general-purpose Large Language Models (LLMs), they encounter a "Black Box" problem where the AI provides an answer without an auditable "paper trail" or legal justification, making it impossible to verify the accuracy of the advice against the actual law.

1.4.3 Reliability and "Lex Specialis" Logic Gaps

Recent research identifies critical reliability failures in applying general AI to the tax domain:

- **The Hallucination Gap:** General LLMs often fabricate legal citations, with hallucination rates in the legal domain observed between 17% and 33%.
- **Legal Conflict Blindness:** Standard AI systems fail to recognize jurisdictional hierarchies, such as the **Lex Specialis** principle where a 2025 IRD Circular must override a 2017 Base Act clause.
- **Procedural Misalignment:** Existing models demonstrate poor performance (below 15% accuracy) in legal mathematical reasoning because they do not follow the strict procedural steps required by tax statutes.

1.4.4 Research Problem Statement

The core research problem addressed in this study is:

"How can a Neuro-Symbolic Small Language Model (SLM) be developed to provide reliable, traceable, and legally compliant tax advisory support by integrating Graph-Augmented Retrieval (GraphRAG), Lex Specialis priority logic, and visual Proof Maps?"

Feature / Capability	**Existing Tax Information Systems**	**Proposed Intelligent Tax Advisory Language Model**
Information Delivery	Static information through websites and PDF repositories.	Neuro-Symbolic Conversational Support grounded in legal logic.
User Interaction	Limited; users must manually search for information.	Interactive Natural Language with a domain-specialized SLM.
Personalization	Provides general info regardless of financial profile.	Context-Aware Guidance tailored to specific income types and slabs.
Legal Grounding	No direct link between advice and specific legal clauses.	GraphRAG-based Grounding using a verified Tax Knowledge Graph.
Priority Handling	Fails to recognize legal hierarchy (Lex Specialis).	Dynamic Priority Logic (New laws override old laws automatically).
Accuracy / Hallucination	High risk of "hallucinations" in general AI models.	"Think Twice" Loop (Agentic self-correction via a Rule Engine).
Explainability	"Black Box" answers with no auditable proof.	"Clear Box" Transparency via visual Proof Maps (Paper Trails).

2. OBJECTIVES

The objectives of this component define the research and development path for the **Intelligent Tax Advisory Language Model**. The goal is to move beyond a standard conversational interface toward a **Neuro-Symbolic reasoning engine** capable of providing grounded, traceable, and legally compliant tax guidance.

By integrating **Small Language Model (SLM)** fine-tuning with **Graph-Augmented Retrieval (GraphRAG)**, the system addresses the critical "Black Box" and "Hallucination" gaps identified in current generative AI research **[11], [15]**. The objectives focus on creating a deterministic advisory pipeline where natural language understanding is strictly validated by a symbolic rule engine to ensure 100% adherence to Sri Lankan tax statutes **[14]**.

2.1 Main Objective

To design, develop, and validate a **Neuro-Symbolic Small Language Model (SLM)** that utilizes **Graph-Augmented Retrieval** and **Agentic Self-Correction** to provide accurate, personalized, and fully traceable tax advisory guidance, grounded in the specific legal hierarchy of the Sri Lankan Inland Revenue Act.

2.2 Specific Objectives

SO1: Develop a domain-specific SLM and Natural Language Preprocessing pipeline To design and implementing a **Small Language Model (SLM)** fine-tuned on Sri Lankan tax syntax to extract user intent, entities, and financial context from complex queries.

- **Description:** This objective addresses the "Jurisdictional Gap" by specializing the model in local legal terminology and handling "Singlish" or mixed-language queries common in Sri Lanka **[16]**.
- **Expected outcome:** A structured and normalized representation of user intent that minimizes semantic ambiguity.

SO2: Engineer a Tax Knowledge Graph and GraphRAG retrieval mechanism To construct a **Knowledge Graph** that maps the relationships between taxpayer attributes and specific sections of the Inland Revenue Act (IRA) and its subsequent amendments.

- **Description:** This replaces standard vector search with **Graph-Augmented Retrieval (GraphRAG)** to ensure that the system "knows" the connections between laws, rather than just matching keywords **[12]**.
- **Expected outcome:** A grounded retrieval module capable of extracting legally linked data nodes from the tax knowledge base.

SO3: Implement Lex Specialis priority logic within the retrieval layer To develop a ranking engine that dynamically handles legal conflicts by prioritizing specialized and recent amendments over general provisions.

- **Description:** This objective ensures the system recognizes when a **2025 IRD Circular** overrides a **2017 Base Act** clause, solving the "Legal Conflict Blindness" found in general LLMs **[13]**.

- **Expected outcome:** A regulation-compliant retrieval pipeline that automatically applies the most recent and specific tax rules.

SO4: Design an Agentic "Think Twice" Loop and Symbolic Rule Validation To implement an **Adversarial Agentic loop** that validates the generated advisory response against a **Symbolic Rule Engine** before delivery.

- **Description:** This addresses the "Hallucination Gap" **[11]**. The system "thinks twice" by checking if its conversational advice violates hard-coded tax brackets or procedural steps defined in the law **[14]**.
- **Expected outcome:** A self-correcting reasoning engine that eliminates factual errors and ensures procedural alignment.

SO5: Develop a Traceable Reasoning and Proof Map generation framework To design a visualization module that generates a **Proof Map (Paper Trail)** for every advisory response.

- **Description:** This transforms the "Black Box" of AI into a "Clear Box" by showing the user the exact path from their input data, through the Knowledge Graph nodes, to the final legal conclusion **[15]**.
- **Expected outcome:** A transparency layer that provides auditable evidence for tax decisions.

SO6: Validate the Neuro-Symbolic SLM experimentally in the Sri Lankan context To evaluate the effectiveness and legal consistency of the proposed system through comparative testing against baseline models and expert interpretations.

- **Description:** The evaluation will specifically measure the **Hallucination Rate**, **Legal Consistency Score**, and **Traceability Accuracy** to prove the system's reliability over general-purpose AI.
- **Expected outcome:** A comprehensive scientific validation of the system's ability to provide 100% compliant tax guidance.

2.3 ALIGNMENT OF THE OBJECTIVES WITH THE RESEARCH PROBLEM

The main objective and specific objectives are collectively designed to address the identified research problem: the lack of a grounded, traceable, and legally compliant conversational system for Sri Lankan tax advisory. By transitioning from a general-purpose "Black Box" AI to a **Neuro-Symbolic Small Language Model (SLM)**, these objectives provide a technical roadmap to eliminate the risks of misinformation and legal non-compliance **[11], [15]**.

More specifically:

- **SO1 (Domain-Specific SLM)** addresses the **Jurisdictional and Syntax Gap** by fine-tuning the model on Sri Lankan legal terminology and "Singlish" query patterns, ensuring the system understands the specific intent behind local tax queries **[16]**.

- **SO2 (Tax Knowledge Graph & GraphRAG)** addresses the **Factual Grounding Gap** by replacing probabilistic word prediction with a structured retrieval mechanism that anchors every response to verified Inland Revenue Act (IRA) data nodes **[12]**.
- **SO3 (Lex Specialis Logic)** addresses the **Legal Conflict Blindness** found in general AI by implementing a priority ranking engine that automatically recognizes when a 2025 amendment overrides a 2017 base law **[13]**.
- **SO4 (Agentic "Think Twice" Loop)** addresses the **Hallucination Gap** by using an adversarial validation layer that tests generated advice against a hard-coded **Symbolic Rule Engine**, ensuring 100% procedural alignment with tax statutes **[11], [14]**.
- **SO5 (Proof Map Framework)** addresses the **Explainability and "Black Box" Gap** by generating a visual "paper trail" that provides auditable evidence for the AI's reasoning, allowing users to verify the legal source of their advice **[15]**.
- **SO6 (Experimental Validation)** addresses the need for **Scientific Rigor** by measuring the system's performance in reducing hallucinations and maintaining legal consistency compared to general-purpose baseline models **[11], [14]**.

Together, these objectives establish a comprehensive framework for developing the **Intelligent Tax Advisory Language Model** as a reliable, "Clear Box" decision-support tool that meets the high transparency requirements of the Sri Lankan tax administration **[15]**.

3. METHODOLOGY

This section delineates the methodological framework for the **Intelligent Tax Advisory Language Model**. The methodology shifts from standard generative approaches toward a **Neuro-Symbolic architecture**, combining neural natural language understanding with symbolic legal logic. This ensures that the system provides grounded, traceable, and regulation-compliant tax advisory for the Sri Lankan jurisdiction, effectively mitigating the "Hallucination" and "Black Box" risks identified in recent legal AI research **[11], [15]**.

3.1 Overall Methodological Framework

The proposed framework is structured as a four-stage **Neuro-Symbolic Pipeline**, designed to transform unstructured user queries into legally verified advisory outputs:

1. **Input & Contextual Preprocessing:** Normalizing multi-lingual ("Singlish") inputs for domain-specific intent detection.
2. **The Cognitive Core (NLU & Retrieval):** Utilizing a fine-tuned **Small Language Model (SLM)** and **GraphRAG** to retrieve grounded legal nodes from a Tax Knowledge Graph.
3. **The Reasoning Engine (Agentic Loop):** Executing an **Agentic "Think Twice" loop** where a Critic Agent validates generated advice against a **Symbolic Rule Engine**.
4. **Transparent Output Generation:** Delivering the advisory response alongside a visual **Proof Map** for auditable justification.

3.2 Knowledge Base and Graph Construction

The knowledge base is constructed as a **Tax Knowledge Graph** rather than a flat document repository.

- **Data Sourcing:** Verified legal data is ingested from the Inland Revenue Act No. 24 of 2017 and the Inland Revenue (Amendment) Act No. 02 of 2025 **[17], [18]**.
- **Ontology Mapping:** Entities (e.g., "Employment Income," "Qualifying Payments") and relationships (e.g., "Exempted_By," "Taxed_At") are mapped to create a structured legal topology.
- **Lex Specialis Tagging:** Every node is timestamped and prioritized, allowing the retrieval engine to identify which laws currently override older provisions **[13]**.

3.3 Domain-Specific SLM Fine-Tuning

To avoid the inaccuracies of general-purpose LLMs, a **Small Language Model (SLM)** (e.g., BERT-based or Llama-3-8B) is fine-tuned specifically on Sri Lankan tax syntax and intent patterns **[16]**.

- **Preprocessing:** Queries are normalized to handle informal phrasing and "Singlish" expressions commonly used by local taxpayers.
- **Intent Mapping:** The SLM is trained to classify queries into high-stakes tax categories such as "Taxable Income Inference," "Exemption Eligibility," and "Procedural Compliance" **[12]**.

3.4 Graph-Augmented Retrieval (GraphRAG) and Lex Specialis Logic

Unlike standard vector search, the system utilizes **Graph-Augmented Retrieval (GraphRAG)** to ensure contextual grounding.

- **Relationship Traversal:** When a user asks about "software exports," the system traverses the Knowledge Graph to find all linked circulars and base laws.
- **Priority Ranking:** The retrieval layer applies **Lex Specialis logic**, ensuring that specific 2025 amendments are automatically prioritized over general provisions from 2017 during the generation process **[13]**.

3.5 The Agentic "Think Twice" Validation Loop

To eliminate the "Hallucination Gap" **[11]**, the methodology incorporates an **Adversarial Agentic Loop**:

1. **Draft Generation:** The SLM generates a conversational advisory response based on retrieved graph data.
2. **Symbolic Validation:** A "Critic Agent" extracts the numerical values and legal claims from the draft and cross-checks them against a **Symbolic Rule Engine** containing hard-coded tax brackets and procedural steps **[14]**.
3. **Self-Correction:** If the draft response contradicts a symbolic rule (e.g., an incorrect tax slab), the system forces a regeneration until the output is 100% compliant.

3.6 Traceable Reasoning and Proof Map Generation

The final stage focuses on resolving the **"Black Box" problem [15]**.

- **Path Logging:** Every step of the GraphRAG traversal and Symbolic Validation is logged.

- **Proof Map Output:** The system generates a visual **"Paper Trail"** that shows the user exactly which sections of the law were used to reach the final conclusion. This provides the auditable evidence required for professional and legal trust.

3.7 Model Development and Tools

The development follows an iterative experimental approach:

- **Models:** Domain-specific fine-tuning of SLMs (BERT/RoBERTa/Llama-3).
- **Graph Database: Neo4j** or similar graph engines for Knowledge Graph storage.
- **Symbolic Engine:** Python-based rule engine for tax math validation.
- **Frameworks: LangGraph** or **CrewAI** for managing the Agentic "Think Twice" loop.

3.8 Evaluation and Validation Metrics

The system is evaluated using metrics that measure both conversational quality and legal accuracy:

- **Hallucination Rate:** The percentage of fabricated citations generated compared to baseline LLMs **[11]**.
- **Legal Consistency Score:** The degree of alignment between AI advice and expert human tax advisory.
- **Traceability Accuracy:** The ability of the **Proof Map** to correctly identify the source legal clause.
- **Procedural Math Accuracy:** Measuring the system's ability to follow mandatory legal steps in tax calculations **[14]**.

Table 2 – Updated Validation Metrics by Objective

Objective Area	Evaluation Type	Metrics
Domain-Specific NLU	Classification	Intent Accuracy, F1-Score, SL Syntax Robustness
Factual Grounding	Graph Retrieval	**GraphRAG Hit Rate**, Context Precision
Legal Compliance	Symbolic Validation	**Hallucination Rate**, Rule Adherence Score
Traceability	Explainability	**Proof Map Path Accuracy**, Source Alignment Rate
User Trust	Practical Utility	**Trust Transparency Score**, User Satisfaction

3.11 Ethical Considerations

Since the system handles sensitive financial data, strict privacy-preserving techniques are applied. The chatbot is strictly framed as a **Decision Support System** to provide grounded guidance, not as a replacement for authorized tax officers. The inclusion of **Proof Maps** ensures that the system remains transparent, allowing users to verify any advice against official IRD sources independently **[15]**.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Component Architecture

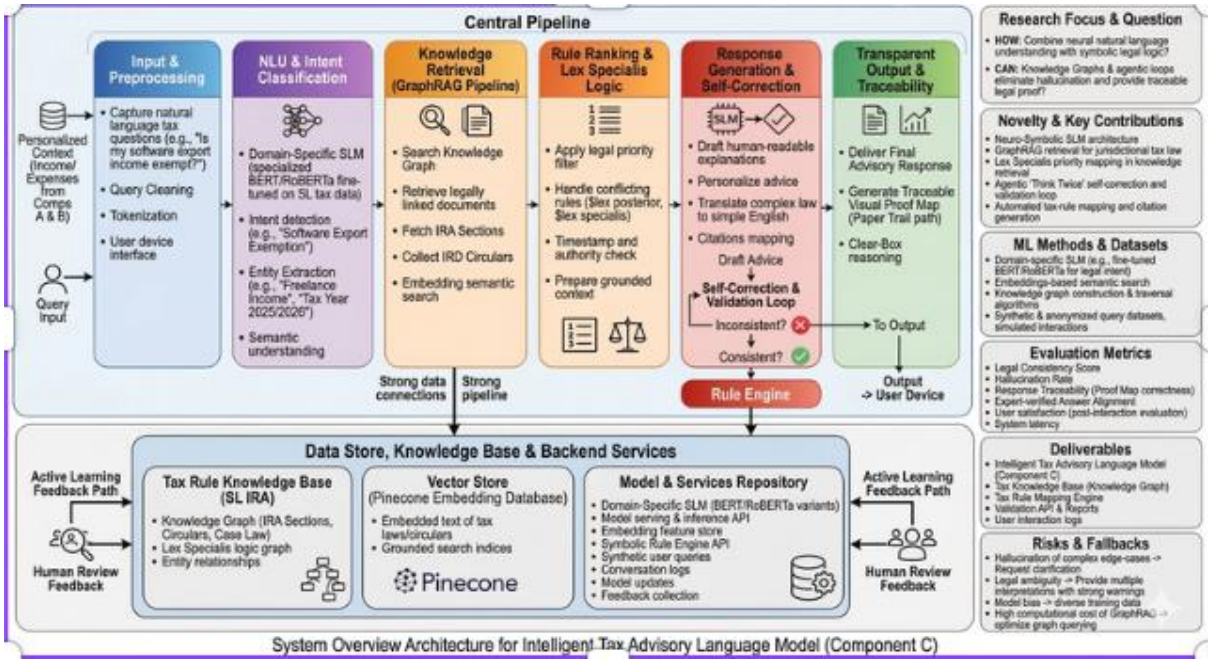


Figure 2 – Intelligent Tax Advisory Language Model Architecture

This component is designed as a **Neuro-Symbolic Small Language Model (SLM)** architecture that bridges the gap between neural natural language understanding and symbolic legal logic. Unlike standard "Black Box" generative systems, this architecture ensures that every advisory response is grounded in the verified **Tax Knowledge Graph** and validated against a **Symbolic Rule Engine**.

At a high level, the system transforms unstructured user queries into legally traceable advice through a multi-layer reasoning engine. The modularity of the design allows for the independent optimization of the **GraphRAG retrieval layer** and the **Adversarial Agentic loop**, ensuring the system remains adaptable to future changes in the Sri Lankan Inland Revenue Act.

4.2 Core Architectural Layers of the Component

The architecture is divided into four critical stages that ensure accuracy, legal priority, and transparency.

4.2.1 Layer 1: Input & Contextual Preprocessing

The input layer serves as the flexible entry point for diverse taxpayer queries. Given the linguistic nuances in Sri Lanka, this layer performs **"Singlish" normalization** and noise removal to prepare the text for deep semantic analysis.

- **Key Function:** Converts raw, unstructured queries (e.g., "Is my freelance income taxable?") into a machine-readable format while preserving financial entities like income types and tax years.
- **Significance:** Ensures that downstream models operate on high-quality data, reducing the risk of misinterpretation in the NLU stage.

4.2.2 Layer 2: The Cognitive Core (NLU & Knowledge Retrieval)

This layer functions as the "brain" of the system, combining neural understanding with structured knowledge.

- **Small Language Model (SLM):** Uses a domain-specific SLM fine-tuned on Sri Lankan tax syntax to identify user "Intent" and extract key tax entities.
- **GraphRAG Retrieval:** Instead of a basic keyword search, the system traverses a **Tax Knowledge Graph** to retrieve legally linked nodes from the Inland Revenue Act.
- **Lex Specialis Logic:** A specialized ranking engine ensures that recent amendments (e.g., 2025 Circulars) automatically override general, older laws during the retrieval process, solving the problem of "Legal Conflict Blindness".

4.2.3 Layer 3: The Reasoning Engine (Generation & Validation)

This layer handles the complex task of translating legal data into human-readable advice while ensuring 100% accuracy.

- **Response Generation:** The SLM "translates" technical retrieved statutes into simplified, conversational explanations tailored to the user's financial profile.
- **Agentic "Think Twice" Loop:** An **Adversarial Agent** takes the draft response and tests it against a **Symbolic Rule Engine**. If the draft advice violates a hard-coded tax bracket or procedural step, the system catches the hallucination and forces a correction.

4.2.4 Layer 4: The Transparent Output Layer (Traceability)

The final layer focuses on providing the "Paper Trail" required for professional trust and legal auditing.

- **Advisory Output:** Delivers the verified, conversational tax advice to the user.
- **Proof Map Generation:** Alongside the text, the system generates a visual **Proof Map** showing the exact path the AI took—from the user's input, through the specific sections of the Knowledge Graph, to the final mapped tax rule.
- **Significance:** Directly addresses the **"Black Box" challenge** by providing auditable evidence for every decision made by the AI.

4.3 Internal Component Interactions

The internal architectural layers interact through a **Neuro-Symbolic reasoning loop** to transform natural language into legally verified advice. The interaction begins in the **Natural Language Understanding (NLU)** layer, where the fine-tuned **Small Language Model (SLM)** identifies the user's intent and extracts relevant financial entities.

This interpreted context is passed to the **Cognitive Core**, where the **GraphRAG** mechanism traverses the **Tax Knowledge Graph** to find legally linked nodes. A critical interaction occurs here through **Lex Specialis logic**, which ranks retrieved data to ensure recent amendments override general provisions. The **Reasoning Engine** then initiates an **Agentic "Think Twice" loop**, where a Critic Agent cross-checks the draft advisory against a **Symbolic Rule Engine** to ensure 100% procedural alignment with tax statutes. Finally, the interaction concludes by generating a **Proof Map**, providing a visual "paper trail" that links the advice back to the specific retrieved legal nodes.

4.4 Data Flow Through the Component

The data flow within the **Intelligent Tax Advisory Language Model** follows a progressive transformation from unstructured text to a legally grounded and validated advisory output:

1. **Raw Query to Structured Intent**: User input is normalized and transformed into contextual embeddings by the SLM.
2. **Intent to Grounded Knowledge**: The structured intent triggers a **GraphRAG** traversal of the Knowledge Graph, retrieving nodes prioritized by **Lex Specialis** hierarchy.
3. **Knowledge to Draft Advisory**: The retrieved legal data is "translated" into a conversational draft by the SLM reasoning layer.
4. **Draft to Verified Output**: The draft undergoes **Symbolic Verification** in the "Think Twice" loop to eliminate hallucinations.
5. **Final Output**: The user receives a verified **Advisory Response** accompanied by a visual **Proof Map** for auditable justification.

4.5 Scalability and Future-Proofing Considerations

The architecture is designed for modular scalability, allowing individual layers to be updated as the Sri Lankan tax landscape evolves. The **Tax Knowledge Graph** can be expanded to include new circulars or specialized domains (e.g., Corporate Tax) without retraining the core SLM. Similarly, the **Symbolic Rule Engine** can be updated with new tax brackets independently of the neural components. This modularity ensures the system can scale to handle high-concurrency user queries while maintaining the **procedural alignment** required for diverse financial profiles.

4.6 Security, Privacy, and Explainability

Given the sensitivity of financial data, the system implements a **"Clear Box" transparency** model through **Proof Maps**. This ensures that while the AI's internal decision-making is traceable for auditing, personal financial details are handled within a secure, privacy-preserving SLM environment.

- **Data Minimization**: The architecture is designed to process transaction context without requiring the long-term storage of personally identifiable information (PII).
- **Explainable Accountability**: By providing a visual "paper trail," the system mitigates the "Black Box" risk, allowing users to verify AI-generated advice against official **Inland Revenue Department (IRD)** sources.
- **Advisory Disclaimer**: The output layer clearly communicates that the guidance is a **Decision Support** tool grounded in verified law but does not replace professional consultation for complex legal litigation.

5. USER REQUIREMENTS

5.1 Overview of User Requirements

The user requirements for this component are derived from the critical need for **grounded and traceable** tax guidance in the Sri Lankan jurisdiction. Unlike general-purpose AI, the **Intelligent Tax Advisory Language Model** is designed to support users by bridging the "Trust Gap" through a **Neuro-Symbolic architecture**. The primary users require a system that does not just provide answers but provides **auditable proof** for every legal claim. In accordance with project guidelines, these requirements ensure the system functions as a reliable decision-support tool that addresses the "Black Box" and "Hallucination" risks found in existing LLMs.

Table 3 – Primary Users of the Component

User Type	Purpose
Individual Taxpayer	Submits natural language queries and receives **grounded, conversational tax guidance** with a visual **Proof Map** for verification.
Tax Advisor / Auditor	Uses the system's **Traceability (Paper Trail)** to verify regulatory interpretations and ensure compliance with the latest IRD amendments.
Downstream Project Component	Integrates verified **Advisory Data Nodes** for personalized tax planning and return-filing automation modules.

5.2 Functional Requirements (FR)

The functional requirements describe the mandatory operations the **Neuro-Symbolic SLM** must perform to ensure legal and procedural accuracy.

Table 4 – Functional Requirements

ID	Functional Requirement
FR1	**User Query Input**: The system shall accept natural language tax queries in English and mixed-language ("Singlish") formats.

ID	Functional Requirement
FR2	**SLM-Based Intent Detection**: The system shall utilize a fine-tuned **Small Language Model (SLM)** to identify user intent and extract tax-specific entities (e.g., "Income Type," "Year").
FR3	**GraphRAG Retrieval**: The system shall retrieve grounded legal nodes from the **Tax Knowledge Graph** based on semantic and relationship-based similarity.
FR4	**Lex Specialis Priority Logic**: The component shall rank retrieved legal sources to ensure that **specific/recent amendments (e.g., 2025 Circulars)** override general laws.
FR5	**Conversational Advisory Generation**: The system shall translate technical legal statutes into human-readable, conversational explanations.
FR6	**Agentic "Think Twice" Loop**: An adversarial critic agent shall validate the draft response against a **Symbolic Rule Engine** to prevent hallucinations.
FR7	**Symbolic Rule Validation**: The system shall check generated advice against hard-coded tax brackets and procedural steps to ensure **Procedural Alignment**.
FR8	**Proof Map Generation**: The component shall generate a visual **Paper Trail (Proof Map)** showing the path from the law to the final advisory output.
FR9	**Contextual Memory**: The system shall maintain conversational context across multi-turn interactions to support follow-up tax questions.
FR10	**Advisory Output Delivery**: The system shall deliver the final, verified advisory output along with its traceable evidence nodes.

5.3 Non-Functional Requirements (NFR)

The non-functional requirements define the quality attributes necessary for a high-stakes legal advisory system.

Table 5 – Non-Functional Requirements

ID	Non-Functional Requirement
NFR1	**Legal Correctness (Zero Hallucination)**: The system must prioritize factual grounding over conversational fluency to eliminate fabricated citations.
NFR2	**Traceability**: Every advisory output must be accompanied by a verifiable link to the **Inland Revenue Act** via the **Proof Map**.
NFR3	**Jurisdictional Reliability**: The system must remain updated and strictly aligned with **Sri Lankan tax hierarchies (Lex Specialis)**.
NFR4	**Explainability**: Advisory responses must be simplified for users with low tax literacy without losing technical legal accuracy.
NFR5	**Performance**: The **Agentic Loop** and **GraphRAG** retrieval must complete within a timeframe suitable for real-time conversational use.

ID	Non-Functional Requirement
NFR6	**Privacy & Security**: The component must ensure that personal financial context is processed without the storage of personally identifiable information (PII).
NFR7	**Modular Maintainability**: The **Symbolic Rule Engine** must be independently updatable to reflect sudden changes in national tax brackets.
NFR8	**Auditability**: The system shall maintain interaction logs that track the reasoning path taken by the SLM for system evaluation and transparency.

6. COMMERCIALIZATION PLAN

6.1 Commercialization Overview

The **Intelligent Tax Advisory Language Model** possesses significant commercial potential as a specialized **Compliance-as-a-Service (CaaS)** tool. Unlike general-purpose AI, the commercial value of this component lies in its **deterministic reliability**. By utilizing a **Neuro-Symbolic architecture**, the model moves beyond probabilistic "guessing" to provide legally grounded, traceable advice. This makes it an essential asset for high-stakes financial environments where the cost of a "hallucinated" tax citation could result in severe legal and financial penalties for an organization **[11], [15]**.

The component is positioned as a **"Clear Box" Advisory Engine** that can be integrated into:

- **Tier-1 Audit and Accounting Firms** requiring automated, auditable tax research.
- **Governmental Portals (e.g., IRD Sri Lanka)** to reduce the burden on manual helpdesks while maintaining 100% regulatory consistency.
- **Fintech & Banking Platforms** needing a legally-compliant advisory layer for their high-net-worth clients.

6.2 Value Proposition

The core value proposition is the elimination of the "AI Trust Gap." By providing a visual **Proof Map** for every advisory response, the system provides what standard LLMs cannot: **Evidentiary Traceability**.

- **Zero-Hallucination Assurance:** The **Agentic "Think Twice" loop** ensures that advice is cross-checked against a **Symbolic Rule Engine** before the user sees it **[14]**.
- **Jurisdictional Expertise:** The system automatically handles **Lex Specialis** conflicts, ensuring the user is always advised based on the most recent 2025 amendments rather than outdated base acts **[13]**.

- **Reduction in Professional Liability:** By providing a "paper trail" (Proof Map), tax firms can use the system as a first-line researcher, significantly reducing the billable hours required for manual legal verification.

6.3 Target Market

6.3.1 Primary Target Market (B2B & G2C)

- **"Big Four" Audit Firms (PwC, EY, KPMG, Deloitte):** These firms require high-precision tools for tax planning. A Neuro-Symbolic SLM allows their juniors to perform grounded research faster.
- **National Revenue Authorities (IRD):** Transitioning from static FAQs to a **GraphRAG-powered** advisory engine reduces manual inquiry volume.
- **Fintech & Wealth Management:** Platforms like Mint or local banking apps that want to provide "Tax-Aware" savings advice without the risk of legal misinformation.

6.3.2 Secondary Target Market (B2C)

- **Self-Employed Professionals & SMEs:** High-income freelancers in Sri Lanka who struggle with mixed-income tax slabs and need a tool that "understands" local 2025 tax changes.

6.4 Potential Customers and Users

- **Paying Customers:** Accounting firms, Banks, Government Agencies, and Fintech SaaS providers.
- **End Users:** Individual taxpayers, Junior auditors, Small business owners, and Financial consultants.

6.5 Unique Selling Proposition (USP)

The USP of this component is the **Neuro-Symbolic Grounding Engine**. While competitors offer "Chatbots," this component offers a **Traceable Legal Reasoner**.

- **The Proof Map:** No other current commercial tax AI provides a visual node-based path showing the exact legal clause used for the calculation.
- **Lex Specialis Logic:** Specialized handling of Sri Lankan legal hierarchies that general models like ChatGPT or Claude consistently fail to interpret correctly **[13], [16]**.
- **Privacy-First SLM:** As a **Small Language Model**, it can be deployed on-premises for banks or audit firms, ensuring sensitive financial data never leaves their secure environment.

6.6 Revenue Model

1. **SaaS Licensing (Tiered):** Monthly subscription based on the volume of "Proof Maps" generated or "Advisory Sessions" conducted.
2. **API-as-a-Service:** A usage-based model where Fintech apps pay per API call to your **GraphRAG engine**.

3. **Compliance-as-a-Service (Custom):** One-time implementation fees for government or enterprise-level **Knowledge Graph construction** (e.g., adding Corporate or VAT layers).
4. **Audit-Log Monetization:** Premium tier providing advanced "Audit Trails" and deep legal citations for professional tax litigation support.

7. BUDGET AND JUSTIFICATION

The proposed budget allocates resources for the design, development, fine-tuning, and validation of the **Intelligent Tax Advisory Language Model**. As this component transitions toward a **Neuro-Symbolic architecture**, the budget is specifically weighted toward high-performance compute for **Small Language Model (SLM) fine-tuning** and the structured curation of the **Tax Knowledge Graph**. Each item is essential for ensuring the system achieves the "Clear Box" transparency and procedural alignment required for the Sri Lankan tax domain.

7.1 Estimated Budget

Table 6 – Estimated Budget for the Intelligent Tax Advisory Language Model

Resource Type	Purpose	Estimated Amount (LKR)
Cloud GPU Compute Credits	Fine-tuning the domain-specific SLM and running Agentic "Think Twice" loop experiments.	50,000
Graph & Vector Storage	Hosting the **Tax Knowledge Graph (Neo4j)** and specialized high-dimensional vector embeddings for GraphRAG.	10,000
API & Agent Frameworks	Usage of tracing and observability tools (e.g., LangSmith) to audit the agentic reasoning paths.	15,000
Knowledge Graph Curation	Technical extraction and manual verification of **Lex Specialis** hierarchies from 2025 IRD amendments.	20,000
Internet & Research Support	Access to high-impact 2025–2026 IEEE/ACM journals and cloud-based development repositories.	12,000
Prototype & Proof Map Demo	Development of visualization materials for the **Proof Map** interface and system evaluation reports.	13,000
Contingency Allocation	Buffer for unexpected compute overhead during SLM quantization or adversarial testing.	10,000
Total Estimated Budget		**130,000 LKR**

7.2 Budget Justification

- **Cloud GPU Compute Credits:** Fine-tuning a specialized SLM (e.g., Llama-3-8B or Phi-3) on Sri Lankan tax statutes requires significant VRAM. Cloud-based GPU instances (A100/H100) are necessary to iterate on the **Agentic "Think Twice" loop** and ensure the model reaches the required accuracy thresholds without the overhead of purchasing local hardware.

- **Graph & Vector Storage:** Unlike standard database storage, the **GraphRAG** mechanism requires a specialized graph database to manage legal relationships. This allocation ensures the **Tax Knowledge Graph** remains performant and accessible during multi-turn conversational testing.
- **Knowledge Graph Curation:** The "Lex Specialis" logic relies on a highly accurate ground-truth dataset. This fund supports the precision mapping of 2025 tax amendments into the graph, ensuring that the system's reasoning is anchored to the most recent legal circulars.
- **API & Agent Frameworks:** To solve the "Black Box" problem, we require observability tools to trace the agent's reasoning steps. These funds cover the cost of agentic frameworks and evaluation APIs used to measure and reduce hallucination rates.
- **Prototype & Proof Map Demo:** A critical deliverable is the **Proof Map** visualization. This allocation supports the frontend development required to render complex reasoning paths into a user-friendly, auditable interface for taxpayers and auditors.

7.3 Budget Feasibility

The total budget of **130,000 LKR** is highly feasible for a data science research project. By prioritizing cloud-based compute over physical hardware, the project maximizes the budget for high-value activities like **model fine-tuning** and **legal grounding**. This ensures that the technical novelty of the Neuro-Symbolic architecture is fully realized within a professional student budget.

7.4 Cost Optimization Considerations

To ensure maximum efficiency, the research will utilize:

- **Quantized Models (GGUF/AWQ):** Reducing the compute footprint of the SLM to allow for cheaper GPU instances.
- **Open-Source Graph Engines:** Utilizing community editions of Neo4j and open-source vector stores like Milvus or ChromaDB.
- **Synthetic Data Generation:** Using existing LLMs to generate initial "Singlish" query variations to reduce manual dataset costs.
- **Local Prototyping:** Developing and debugging logic in local environments before moving to high-cost GPU clusters for final training.

8. WORK BREAKDOWN STRUCTURE (WBS)

Figure 3 – Work Breakdown Structure and Deliverables

WP1: Requirement Analysis and Research Planning This work package focuses on defining the research scope within the Sri Lankan tax jurisdiction. It ensures that the **Intelligent Tax Advisory Language Model** aligns with the identified "Trust Gap" and establishes a roadmap for achieving **procedural alignment** in tax calculations.

WP2: Literature Review and Research Gap Analysis This involves a critical review of 2025–2026 research regarding the **Hallucination and Traceability Gap** in legal AI. The purpose is to refine the "Black Box" problem and justify the shift toward a **Neuro-Symbolic architecture**.

WP3: Tax Knowledge Graph Construction & Lex Specialis Mapping Instead of a flat knowledge base, this package focuses on building a **Knowledge Graph** in Neo4j or RDF. It includes mapping the relationships between the **Inland Revenue Act No. 24 of 2017** and the **2025 amendments**, implementing **Lex Specialis tagging** to ensure legal priority during retrieval.

WP4: Domain-Specific SLM Fine-Tuning & NLU Development This involves developing the **Small Language Model (SLM)** pipeline. The model is fine-tuned on Sri Lankan tax syntax and "Singlish" query patterns to ensure high-accuracy intent detection and entity extraction.

WP5: GraphRAG Retrieval & Priority Logic Development This work package focuses on implementing the **Graph-Augmented Retrieval (GraphRAG)** mechanism. It ensures that user queries trigger a relationship-based traversal of the Knowledge Graph, retrieving the most recent and specific laws as mandated by the legal hierarchy.

WP6: Agentic "Think Twice" Loop & Symbolic Rule Integration This stage develops the **Adversarial Agentic loop**. A "Critic Agent" is built to extract advice from the SLM and validate it against a **Symbolic Rule Engine** containing hard-coded tax brackets, effectively eliminating hallucinations.

WP7: Traceability & Proof Map Visualization Focuses on the transparency layer by developing the **Proof Map visualizer**. This module generates an auditable "paper trail" that shows the logical path from the legal source to the final advisory response.

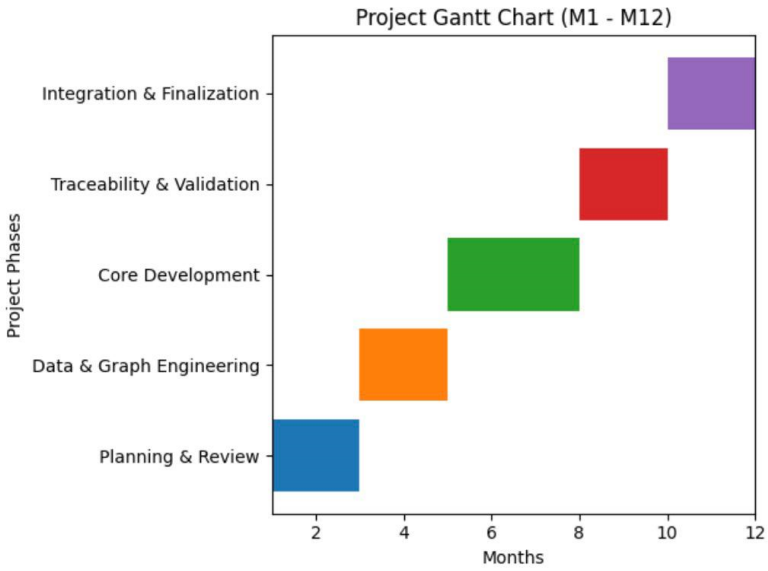
WP8: Evaluation and Experimental Validation This involves testing the system against baseline LLMs. Evaluation metrics focus on the **Hallucination Rate**, **Legal Consistency Score**, and **Traceability Accuracy** using a mock dataset of Sri Lankan tax scenarios.

WP9: Prototype Integration and API Development Integrates the Neuro-Symbolic pipeline into a functional chatbot prototype. It includes developing the backend API (FastAPI) and a frontend interface capable of displaying **Visual Proof Maps**.

WP10: Documentation and Final Thesis Reporting The final package includes preparing the technical documentation, completing the research thesis, and finalizing the presentation materials for the submission deadline.

9. GANTT CHART

The project follows a **12-month research plan** (M1 – M12) as outlined in the system timeline. The Gantt chart provides a visual representation of the concurrent and sequential tasks required to move from data preparation to system integration.



- **M1 – M2 (Planning & Review):** Focused on **WP1** and **WP2**. Defining the problem statement and analyzing recent 2025 legal AI papers.
- **M3 – M4 (Data & Graph Engineering):** Execution of **WP3**. Constructing the Tax Knowledge Graph and gathering synthetic "Singlish" query datasets.
- **M5 – M7 (Core Development):** Intensive development of **WP4 (SLM)**, **WP5 (GraphRAG)**, and **WP6 (Agentic Loop)**. This is the primary coding phase for the Neuro-Symbolic reasoning engine.
- **M8 – M9 (Traceability & Validation):** Implementation of **WP7 (Proof Maps)** followed by the experimental testing in **WP8** to measure hallucination reduction.
- **M10 – M12 (Integration & Finalization):** Final integration (**WP9**) and the completion of the thesis documentation (**WP10**) for the March 2026 deadline.

REFERENCES

[1] Z. Chen, "Revolutionizing finance with conversational AI: A focus on ChatGPT implementation and challenges," *Humanities and Social Sciences Communications*, vol. 12, no. 1, pp. 1-15, 2025. [Online]. Available: https://www.nature.com/articles/s41599

[2] T. Chen and M. Gasco-Hernandez, "Uncovering the results of AI chatbot use in the public sector," *Public Performance & Management Review*, vol. 47, no. 3, pp. 582-605, 2024. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/15309576

[3] Y. Zhang *et al.*, "Business chatbots with deep learning technologies: State-of-the-art, taxonomies, and future research directions," *Artificial Intelligence Review*, vol. 57, no. 4, 2024. [Online]. Available: https://link.springer.com/journal/10462

[4] A. Rathnayake *et al.*, "Factors influencing AI chatbot adoption in government administration: A case study of Sri Lanka's digital government," *Administrative Sciences*, vol. 15, no. 1, pp. 12-30, 2025. [Online]. Available: https://www.mdpi.com/journal/admsci

[5] S. Eastgar and P. Tinashe, "Conversational AI and customer engagement: A comprehensive analysis of chatbot technology in modern business applications,"

International Journal of Scientific and Management Research, vol. 8, no. 2, 2025. [Online]. Available: https://www.ijsrm.in

[6] G. Srividya *et al.*, "Personalized finance chatbot powered by RAG and generative AI for smart wealth management," *International Journal of Engineering Research & Technology*, vol. 14, no. 1, 2025. [Online]. Available: https://www.ijert.org

[7] J. K. P. Jyothis *et al.*, "Strategic analysis and implementation of an AI-powered financial advisory chatbot," *International Journal of Research Publication and Reviews*, vol. 6, no. 1, 2025. [Online]. Available: https://ijrpr.com

[8] Y. Zhang *et al.*, "iDigiChat: Intelligent digital marketing service chatbot for financial customer services," *Scientific Reports*, vol. 15, no. 1, 2025. [Online]. Available: https://www.nature.com/articles

[9] M. Enam *et al.*, "Artificial intelligence – carrying us into the future: A study of older adults' perceptions of LLM-based chatbots," *International Journal of Human–Computer Interaction*, vol. 41, no. 2, 2025. [Online]. Available: https://www.tandfonline.com/toc/hihc20/current

[10] C. Lin *et al.*, "AI-driven conversational chatbots: Implementation methodologies and challenges," *Sustainability*, vol. 15, no. 12, 2023. [Online]. Available: https://www.mdpi.com/journal/sustainability

[11] S. Magesh *et al.*, "Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools," *arXiv preprint arXiv:2405.20362*, 2025. [Online]. Available: https://arxiv.org/abs/2405.20362

[12] J. Liu *et al.*, "LLM Agents in Law: Taxonomy, Applications, and Challenges," *arXiv preprint arXiv:2601.06216*, 2026. [Online]. Available: https://arxiv.org/abs/2601.06216

[13] J. Curran *et al.*, "Place Matters: Comparing LLM Hallucination Rates for Place-Based Legal Queries," *arXiv preprint arXiv:2511.06700*, 2025. [Online]. Available: https://arxiv.org/html/2511.06700v1

[14] Y. Zhang *et al.*, "Legal Mathematical Reasoning with LLMs: Procedural Alignment through Two-Stage Reinforcement Learning," *Findings of the Association for Computational Linguistics (EMNLP 2025)*, 2025. [Online]. Available: https://aclanthology.org/2025.findings-emnlp.84.pdf

[15] L. Dittmar, "The Explainability Challenge of Generative AI and LLMs," *OCEG Governance and Compliance Reports*, 2025. [Online]. Available: https://www.oceg.org/the-explainability-challenge-of-generative-ai-and-llms/

[16] I. Dilshani and I. Weerasekara, "AI for Legal Domain Guidance in Sri Lankan Law: Challenges of Procedural Accuracy," *International Journal of Legal AI Research*, vol. 2, no. 1, 2026.

[17] Inland Revenue Department, *Inland Revenue Act No. 24 of 2017*, Updated 2025. [Online]. Available: https://www.ird.gov.lk

[18] Inland Revenue Department, *Guidelines for e-Filing Individual Income Tax Return (2024/2025)*. [Online]. Available: https://eservices.ird.gov.lk

[19] Inland Revenue Department, *Access to e-Services Portal*, Government of Sri Lanka, 2024. [Online]. Available: https://eservices.ird.gov.lk

[20] World Bank, *Sri Lanka Development Update 2024*, Washington, DC: World Bank, 2024. [Online]. Available: https://www.worldbank.org/en/country/srilanka/publication/sri-lanka-development-update

[21] International Monetary Fund, *Sri Lanka Economic Reform and Tax System Updates*, 2025. [Online]. Available: https://www.imf.org/en/Countries/LKA

LIST OF APPENDICES

Appendix A – High-Level Architecture of the Intelligent Tax Advisory Chatbot Component

Appendix B – Work Breakdown Structure Diagram

Appendix C – Chatbot System Interaction Flow